

ME 210 — Mechanical Engineering Drawing

Laboratory · Fall 2026 — Term 261

Course code	ME 210
Component	Laboratory
Credit hours	1 (0-3-0)
Prerequisite	None
Textbook	Giesecke et al., Technical Drawing with Engineering Graphics, Pearson.

Course description

Hands-on laboratory in engineering graphics: sketching, orthographic projection, sectional and auxiliary views, dimensioning and tolerancing, assembly drawings, and solid modelling with a parametric CAD package.

Learning outcomes

- Produce freehand pictorial and orthographic sketches of machine parts.
- Construct sectional and auxiliary views to standard conventions.
- Dimension and tolerance a drawing so that it can be manufactured.
- Build parametric solid models and generate drawings from them.
- Assemble parts into a working model and produce a bill of materials.

Weekly plan

Week	Topic
1	Lab safety, drawing standards and lettering
2-3	Freehand sketching and pictorial views
4-5	Orthographic projection
6	Sectional views
7	Auxiliary views
8	Mid-term practical assessment
9-10	Dimensioning and introduction to tolerancing
11-12	Parametric solid modelling
13	Assembly drawings and bill of materials
14-15	Term project and final practical examination



Assessment

Component	Weight
Weekly drawing sheets	35%
CAD exercises	20%
Term project	20%
Final practical examination	25%